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Shaping responsible GenAI use in research through AI literacy-oriented guidelines: insights from postgraduate students

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Abstract

The widespread adoption of generative Artificial Intelligence (GenAI) among researchers has intensified the need for guidelines that safeguard academic integrity and ethical scholarly practice. However, existing GenAI policies tend to focus on teaching, learning and assessment rather than on research, where ethical concerns are considerably more complex. Given this complexity, such guidelines should move beyond rule enforcement to scaffold the AI literacy researchers need when deciding how, when, and whether to use GenAI across the full research process. To address this gap, this study conducted seven focus groups with 28 postgraduate research (PGR) students, who are among the most active adopters of GenAI tools in academic work. It examined how they used GenAI in research, how they interpreted ethically acceptable practice, what concerns they hold, and what institutional support they expected. Drawing on a four-dimensional AI literacy framework, the study analysed PGR students' practices and perceptions through thematic analysis to examine how each dimension of AI literacy was reflected within the research setting. The findings showed that students' awareness of GenAI's capabilities, limitations, and risks corresponded to the "Know & Understand AI" dimension, while their diverse, discipline-specific applications of GenAI across the research workflow reflected the "Use & Apply AI" dimension. Their nuanced boundaries between ethical and unethical use demonstrated the "Evaluate & Create AI" dimension, and concerns about output accuracy, originality, data privacy, and skill degradation highlighted the "AI Ethics" dimension. Building on PGR students' research practices and support expectations, the study proposed researcher-oriented GenAI guidelines that foreground each AI literacy dimension across diverse research tasks, with the aim of informing more applicable policy and sustaining researchers' AI literacy development.

Keywords AI policy, AI literacy, Postgraduate research, Generative AI, Research integrity

Introduction

The growing prevalence of GenAI in research workflows has not been matched by a commensurate expansion of research-specific GenAI guidance. Existing policies primarily target teaching, learning and assessment, emphasising plagiarism and fairness,

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while largely overlooking GenAI use in research contexts, where the core ethical concerns extend beyond plagiarism to include originality of knowledge, authorship, methodological rigour, reproducibility, and compliance with ethical and data protection standards (An et al., 2025; Smith et al., 2025). Even where research-oriented policies exist, they often emphasise high-level principles such as transparency and accountability but remain abstract and difficult to operationalise in day-to-day research practices (Lin, 2025). This abstraction is often attributed to their top-down formulation by policymakers and administrators, and has therefore been criticised for being disconnected from real-world contexts and the lived experiences of researchers, as well as for failing to address concrete research tasks beyond writing, such as literature review, data analysis and result interpretation (An et al., 2025; Lin, 2025).

The lack of practical, actionable guidance on GenAI use in research creates ambiguity around acceptable and unacceptable practices, leaving researchers to interpret appropriate use on their own and increasing the likelihood of unintentional misconduct. This risk is particularly pronounced for PGR students, who constitute both a highly active cohort of GenAI users and emerging researchers still developing ethical reasoning about GenAI use in research, including how to consider scholarly norms, values, risks, and responsibilities when deciding whether a given use of GenAI is morally acceptable or appropriate (Dai et al., 2023; Smith et al., 2025). At the same time, unclear guidance may cause researchers who use GenAI responsibly to experience unnecessary feelings of guilt (Chan, 2025a) or uncertainty, which can undermine confidence, discourage transparent disclosure, or lead to inconsistent and cautious research practices (Lin, 2025). Together, these effects may lead to low-quality, inappropriately AI-generated, or even fraudulent academic work, exposing both researchers and institutions to reputational risk and undermining trust in the wider research enterprise and society.

Guidelines designed solely as regulatory instruments are insufficient for supporting responsible GenAI use in research, where situations are too varied and judgement-dependent for rules alone to anticipate (Lin, 2025). Such guidelines should therefore also function as developmental supports that cultivate researchers' AI literacy over time (Smith et al., 2025). Researchers with stronger AI literacy are better able to interpret and apply rules across complex research situations, while well-designed guidelines can, in turn, scaffold the further development of AI literacy by making responsible GenAI use more actionable and context-sensitive (Haroud & Saqri, 2025).

Drawing on Ng et al. (2021)'s AI literacy framework, this study first investigated how AI literacy was enacted in the research context by examining PGR students' GenAI practices, ethical reasoning, concerns, and expectations for institutional support. These empirical insights informed an initial draft of the guidelines, which was then refined into more comprehensive guidance covering diverse research tasks, with each task addressed through content tied to every dimension of AI literacy.

This study offers three contributions to both theory and practice. First, it extends existing research on GenAI in higher education by shifting attention from teaching, learning, and assessment to the under-examined context of research. Second, the study contextualises AI literacy within research practice by showing how PGR students' GenAI practices and perceptions reflect knowledge, practical, evaluative, and ethical dimensions of AI literacy. Third, the study translates these empirical and theoretical insights into a two-layered framework intended as an illustrative and adaptable basis for supporting

researchers' responsible GenAI use, and the ongoing development of AI literacy across disciplinary and institutional contexts.

Literature review

This review approaches PGR students' GenAI engagement through the lens of AI literacy outlined above. Doing so shifts the question from how PGR students use GenAI to whether they are developing the competencies to use it in ways that are productive for research and responsible toward scholarly standards. Three bodies of literature inform this account: empirical studies of GenAI use in higher education and the ethical concerns it raises ("GenAI use in higher education"–"Ethical concerns surrounding GenAI use" sections); conceptual work on AI literacy and responsible use of GenAI ("AI literacy and responsible GenAI use" section); and the policy landscape in which researchers' practices are situated ("Existing GenAI policies" section). Read together, they show that AI literacy in research remains under-explored, particularly in how researchers enact it across research tasks. Existing guidelines, meanwhile, remain principle-based and weighted toward the risks and ethical concerns associated with GenAI, addressing the "Know & Understand AI" and "AI Ethics" dimensions, while offering little practical guidance for "Use & Apply AI" or "Evaluate & Create AI". This imbalance is increasingly out of step with the rapid adoption of GenAI in research and hinders the development of researchers' AI literacy as an integrated capacity.

GenAI use in higher education

Existing studies have consistently reported high levels of GenAI adoption in higher education across national contexts and academic tasks (Sousa & Cardoso, 2025; Xiao et al., 2025). Undergraduate students appeared to use GenAI primarily to support course-related learning activities, such as completing course assignments (Pedro & Tatiana, 2025; Xiao et al., 2025). In contrast, postgraduate students, including doctoral and master's students, tended to use GenAI for more specialised and research-oriented tasks, such as conducting literature reviews (Deschenes & McMahon, 2024; Yeung et al., 2025), interpreting data (Dai et al., 2023; Thuy & Tien, 2025), and assisting with academic writing (Dai et al., 2023; Pedro & Tatiana, 2025). Evidence further suggested that postgraduate students engaged with GenAI more frequently and intensively than undergraduates (Yeung, 2025).

While these studies establish that postgraduate students use GenAI across a range of research-related tasks, they offer limited insight into how such use unfolds in practice, such as researchers' perceptions of GenAI's capabilities and limitations (the "Know & Understand AI" dimension), and the specific ways they integrate GenAI into their workflows (the "Use & Apply AI" dimension). Without this finer-grained understand, GenAI guidance risks remaining abstract and misaligned with researchers' actual needs (English et al., 2025; Smith et al., 2025).

Ethical concerns surrounding GenAI use

Prior research has identified a range of concerns associated with university students' use of GenAI, particularly regarding accuracy and reliability of AI outputs (Deschenes & McMahon, 2024; Sousa & Cardoso, 2025), data privacy (Pedro & Tatiana, 2025), plagiarism (Chan, 2025b), and over-reliance (Chan & Hu, 2023). These concerns become more

complex in research contexts where GenAI use is closely tied to knowledge production rather than to learning or assessment alone (Resnik & Hosseini, 2025).

Accuracy and reliability problems, for example, refer to cases in which AI-generated outputs are factually incorrect, fabricated, outdated, inconsistent, unverifiable, or not aligned with accepted scholarly evidence. In research contexts, such problems do not merely result in incorrect answers, but may distort literature reviews, misinform research design, compromise data interpretation, or lead to unsupported claims in scholarly outputs (An et al., 2025). Data privacy concerns are also heightened in human-centred research areas which involve human participants, lived experiences, or sensitive qualitative data, making the input of such information into GenAI systems a potential breach of research ethics, consent conditions, or data protection requirements (English et al., 2025). Plagiarism concerns extend beyond copying AI-generated text to more complex questions about authorship (Sabbaghan & Eaton, 2025). In research contexts, authorship is not merely a matter of textual production, but is closely tied to intellectual contribution, accountability, and the capacity to take responsibility for the claims, methods, and interpretations presented in scholarly work. Over-reliance on GenAI is similarly more consequential in research because it may weaken researchers' critical thinking and analytical independence, particularly if they depend on AI-generated outputs without sufficiently questioning, interpreting, or defending the scholarly decisions involved (Chan, 2026; Sabbaghan & Eaton, 2025).

Despite these complex ethical implications, much of the existing literature stops at identifying students' concerns, without examining how researchers themselves navigate the boundary conditions and grey zones that this complexity creates in practice (the "Evaluate & Create AI" and the "AI Ethics" dimensions). Without insight into their situated ethical reasoning, principles for responsible GenAI use risk remaining too abstract to engage with the decisions they are meant to inform.

AI literacy and responsible GenAI use

Research workflows consist of many small, discipline-specific tasks, and the ethical implications of GenAI use may vary according to the purpose of use, the stage of research, the sensitivity of the data involved, and the degree of human intellectual contribution (Cheng et al., 2025). Disciplinary norms are central to these judgements because they shape what counts as acceptable evidence, appropriate method, legitimate authorship, sufficient originality, and responsible interpretation within a field (Smith et al., 2025). For example, the ethical acceptability of using GenAI for coding, writing, data analysis, or visualisation may differ depending on whether these activities are considered core intellectual contributions or supporting technical tasks in a given discipline. Although guidelines can articulate important principles for responsible GenAI use, they cannot fully anticipate every task-specific and context-dependent decision researchers encounter in everyday research practice. Therefore, responsible GenAI use depends not only on the existence of rules, but also on researchers' AI literacy to interpret, adapt, and apply guidance across complex and varied research situations.

However, AI literacy has not yet been sufficiently defined or contextualised in relation to research practice. Ng et al. (2021)'s four-dimensional AI literacy framework provides a useful lens for addressing this gap because it conceptualises AI literacy as an integrated set of knowledge, practical, evaluative, and ethical capacities: "Know & Understand AI",

"Use & Apply AI", "Evaluate & Create AI", and "AI Ethics". Each dimension can be meaningfully interpreted in the research context. "Know & Understand AI" refers to researchers' awareness of GenAI's capabilities, limitations, and risks. "Use & Apply AI" concerns researchers' ability to employ GenAI appropriately across research tasks. "Evaluate & Create AI" involves judging the accuracy, originality, legitimacy, and scholarly value of AI-mediated outputs, while ensuring that human intellectual contribution remains central. "AI Ethics" refers to researchers' capacity to reason about ethical issues including authorship, originality, privacy, accountability, and over-reliance.

Despite the importance of AI literacy for responsible engagement with AI (Chan & Colloton, 2024; Ng et al., 2021), little is known about how dimensions of AI literacy are enacted in research contexts. Examining how PGR students currently understand GenAI's capabilities and limitations, apply it to research tasks, evaluate AI-mediated outputs, and reason about ethical risks can inform guidelines that move beyond rule-setting to support the sustained development of researchers' AI literacy.

Existing GenAI policies

If AI literacy is to be developed rather than assumed, the institutional environment in which researchers work matters as much as the competencies they bring to it. Existing GenAI policies predominantly focus on teaching, learning and assessment, with substantially less attention given to research practice. A review of leading universities in the United States found that only a small minority had issued guidelines specifically addressing researchers' use of GenAI (An et al., 2025). The authors of the present study also conducted a small-scale review of publicly available GenAI guidance at the eight University Grants Committee-funded universities in Hong Kong. While many universities provide general guidance on GenAI use in teaching, learning, or assessment, explicit research-oriented guidance is less consistently available. Where research-specific guidance exists, it commonly addresses issues such as documentation, data security, output verification, accountability, and publication. This indicates an important move toward more research-specific GenAI governance. At the same time, the wider policy landscape still points to a need for further task-sensitive support that can help researchers translate general principles into situated decisions across day-to-day research practices.

Guidance issued by publication ethics organisations, editorial bodies, and publishers, including the Committee on Publication Ethics (COPE) (Committee on Publication Ethics, 2023), the International Committee of Medical Journal Editors (ICMJE) (International Committee of Medical Journal Editors, 2026), Springer Nature (Springer Nature, n.d.), and Wiley (Wiley, n.d.), provides important expectations around authorship, accountability, disclosure, confidentiality, and verification (Resnik & Hosseini, 2025; Smith et al., 2025). Wiley's guidance is particularly useful in moving beyond general principles by offering practical advice on several research and writing activities, including literature review, drafting and editing manuscripts, and data visualisation. However, as publisher guidance, it remains primarily oriented toward publication integrity, manuscript preparation, and submission-stage compliance. It therefore needs to be complemented by researcher-facing and developmental support that helps PGR students exercise situated judgement across the broader research workflow and develop AI literacy across knowledge, application, evaluation, and ethical dimensions.

Recent scholarly work has begun to propose frameworks for responsible GenAI use in research at both individual and institutional levels (Resnik & Hosseini, 2025; Smith et al., 2025). Although these frameworks provide valuable ethical direction, they remain largely principle-based and offer limited operational guidance for applying responsible-use principles across concrete research workflows.

Read through the lens of AI literacy, the current policy landscape supports the "AI Ethics" dimension at the level of principle but provides little scaffolding for "Know & Understand", "Use & Apply", or "Evaluate & Create", the developmental capacities that enable researchers to operationalise those principles. The absence of such scaffolding leaves AI literacy framed as a property of the individual researcher, decoupled from the institutional structures that should support its development.

Research questions

Taken together, the gaps identified across "GenAI use in higher education"–"Existing GenAI policies" sections motivated this empirical work, which examined how PGR students used GenAI in research, how they reasoned about its ethical use, and what institutional support they considered necessary, with the aim of translating these insights into operational, AI-literacy-informed guidance. Specifically, this study was guided by the following four research questions:

RQ1: How do PGR students use GenAI in research?

RQ2: How do PGR students conceptualise ethically acceptable and unacceptable uses of GenAI in research?

RQ3: What ethical concerns do PGR students hold regarding the use of GenAI in research?

RQ4: What forms of institutional support do PGR students believe are necessary to enable responsible use of GenAI in research?

Methodology

Context and participants

The study was conducted at the University of Hong Kong. Its aim was to generate contextually grounded insights into PGR students' GenAI practices, ethical reasoning, and support needs in research, rather than to produce statistically generalisable findings. The targeted participants, i.e., PGR students including MPhil and PhD students, were recruited by sending invitation emails. A total of 28 students from various faculties agreed to participate, spanning from first-year MPhil students to fourth-year PhD students (see Table 1). These participants were grouped into seven focus groups based on broad disciplinary domains and availability, with each group comprising three to six participants. Six groups were formed around shared disciplinary domains (Arts and Design; Business and Data; Health and Medicine; Science and Engineering; and two Social

Table 1 Distribution of participants by year of study

Year of study	Number of participants
MPhil Year 1	3
PhD Year 1	7
PhD Year 2	10
PhD Year 3	5
PhD Year 4	3
Total	28

Sciences and Education groups). One focus group comprised participants from multiple faculties and was formed due to scheduling availability rather than disciplinary similarity. The total number of participants and the faculty composition of each focus group are presented in Table 2.

Focus group interview

Seven semi-structured focus group interviews were conducted using Zoom meetings, each lasting approximately 60 min. All interviews were conducted in English, the primary academic language of the participants' research programmes. A list of questions was used to ensure consistency across interviews while allowing flexibility to follow issues raised by participants. Key areas included: whether and how students use GenAI in their research; which tools they use and for what stages or purposes in the research workflow; their views on when GenAI use is acceptable or unacceptable for research students, in general and in their discipline; ethical and practical concerns (e.g., accuracy, originality, data sensitivity, misinterpretation, overreliance); experiences with institutional guidance related to GenAI; and their views on what should be included in a responsible AI use policy for PGR students and research. Interview questions are provided in Appendix A.

Interview data collection and analysis

Ethical approval for the study was obtained from the Research Ethics Committee of the University of Hong Kong (Reference No.: EA230295). Informed consent was obtained from all participants for the collection and use of their responses. Five interviews were audio-recorded, while two were documented through written notes at the participants' request. For the two note-based interviews, two researchers were involved in managing the written records: one facilitated the discussion, while the other took detailed notes during the session. Immediately after each interview, the two researchers reviewed and consolidated the notes to improve accuracy and ensure that participants' key points, illustrative examples, and main discussion threads were retained. Audio recordings were transcribed verbatim to facilitate qualitative analysis, and all identifying information was removed and replaced with pseudonyms during transcription.

A thematic analysis was adopted to enable systematic identification and comparison of recurring patterns across participants' accounts (Guest et al., 2012). Coding was conducted inductively, with codes developed from the data while being guided by the research questions. The analysis focused on semantic patterns in participants' accounts, attending to explicitly articulated practices and perceptions.

Table 2 The total number of participants and the faculty composition of each focus group

Focus group	Number of participants	Faculty composition
Arts and design	4	Architecture (1), Arts (3)
Business and data	3	Business and Economics (2), Data Science (1)
Health and medicine	5	Dentistry (3), Medicine (2)
Interdisciplinary	3	Medicine (1), Engineering (1), Business and Economics (1)
Science and engineering	4	Science (3), Engineering (1)
Social sciences and education group 1	6	Social Sciences (3), Education (3)
Social sciences and education group 2	3	Social Sciences (2), Education (1)
Total	28	

Two researchers coded one transcript independently to develop an initial coding framework. They then compared their coding, discussed differences in interpretation, and refined the initial codebook through consensus. The remaining six interviews were then coded independently (three each). During this process, the codebook was iteratively refined as new codes emerged, overlapping codes were merged, and unclear code definitions were clarified. Each coded interview was reviewed by a second researcher to ensure consistency, completeness, and alignment with the developing coding framework. Interpretive disagreements were resolved through discussion until agreement was reached.

After coding was completed, the researchers collaboratively reviewed the full set of codes and grouped related codes into higher-order themes. This process involved examining conceptual connections among codes, comparing patterns across interviews, and considering how codes related to the four research questions. The final thematic structure was developed through iterative discussion among the researchers, with themes refined to ensure that they were internally coherent, distinct from one another, and adequately supported by the data. Given the qualitative and context-specific nature of the study, we did not treat saturation as a numerical endpoint. Instead, we assessed the adequacy of the thematic structure by considering whether the final themes captured recurring patterns across the interviews and provided sufficient coverage of the four research questions. Transcripts were managed in Microsoft Word, and coding was conducted in Microsoft Excel.

Results

GenAI uses by postgraduate research students (RQ1)

Table 3 summarises GenAI uses among PGR students in this study. Nearly all participants (27 out of 28) reported using GenAI across multiple stages of the research process. Overall, GenAI was primarily positioned as a supportive and efficiency-enhancing research assistant rather than a substitute for core intellectual work, with human oversight consistently retained.

Ideation and brainstorming: Six participants described using GenAI during the early stages of research to generate, discuss, refine, or polish emerging research ideas. Participants emphasised that AI-generated ideas were not adopted uncritically as such ideas could be inaccurate or misleading. For example, P4 (Dentistry, PhD Year 1) noted that AI suggestions could be "*misleading*" and "*totally wrong*", explaining that while they would "*discuss initial ideas with AI*", they did not want to "*copy and follow the initial idea from the AI*".

Literature review: GenAI use in literature review activities was reported by nine participants. They described using specialised GenAI-enabled tools, such as Perplexity and NotebookLM, to search for relevant academic literature, gain an overview of published work, and summarise existing studies. In particular, participants emphasised how these tools allowed them to retrieve large volumes of relevant literature quickly through simple keyword inputs. As P25 (Architecture, PhD Year 2) explained:

> *There are already a lot of GenAI tools, particularly for literature review. You can just input the keywords and the key discipline you are working on, and it will throw out, for example, 500 connected papers to you, which really makes the whole process less time-consuming.*

Table 3 GenAI uses by postgraduate research students

Category	Participants (n = 28)	Main uses	Role of GenAI	Human oversight
Ideation & brainstorming	6	Generating and refining research ideas	Catalyst for idea development	Critical evaluation; retained conceptual ownership
Literature review	9	Searching, summarising, synthesising literature	Time-saving support tool	Critical reading and integration remained human-led
Explanation	4	Explaining concepts, methods, derivations	On-demand learning aid	Deeper understanding developed independently
Data processing & preparation	2	Data labelling, pre-/post-processing	Support for repetitive/procedural tasks	Output reviewed and validated by researchers
Programming & coding	14	Code generation, debugging, learning languages	Efficiency and learning support	Testing and verification required
Domain-specific tasks	5	Archival processing, protein modelling, AI integration	Embedded specialised research tool	Applied within researcher-controlled workflows
Academic writing	5	Drafting sections of papers/proposals	Partial drafting support	Clear limits to protect originality
Editing & proofreading	16	Grammar, clarity, style improvement	Post-writing refinement tool	Careful review to preserve meaning
Visual content generation and editing	5	Image/video generation, design exploration	Creative and communication support	Human review implied
Administrative and routine tasks	6	Emails, forms, citations, interview prep	Reducing administrative burden	Human review implied
Translation	1	Mandarin–English translation	Multilingual communication support	Human review implied

Explanation: Four participants reported using GenAI such as ChatGPT, DeepSeek and Gemini to explain concepts, mathematical derivations, or research methods encountered during their research. In these cases, GenAI functioned as an on-demand explanatory resource to support understanding of unfamiliar or complex topics, particularly at an introductory rather than in-depth level. P24 (Business and Economics, PhD Year 2) described turning to Gemini for general overviews of unfamiliar economics methods:

> *I use Gemini the most recently. I find that it's quite useful because I study management strategy, we have some economics method that I hadn't heard of before. So I think it's quite helpful when there's a new concept that I need to understand, but not that deep – only the general idea, the history, or how this method is used.*

Data processing and preparation: A small number of participants reported using GenAI to assist with data processing and preparation tasks, such as pre-processing or post-processing raw data and supporting data labelling. GenAI was positioned as a supporting tool for handling procedural and repetitive aspects of data preparation, particularly in tasks involving large or complex datasets. P8 (Data Science, PhD Year 2), for example, described using GenAI to help label video data collected for research purposes:

> *I may collect some videos as demonstrations of robots doing something, and sometimes I also need to use the ChatGPT or other tools, or some other vision-language*

> *models to label the data – for example, simply telling me what the robot is doing, or what a human is doing in the video.*

Programming and coding: Programming and coding support emerged as one of the most prevalent uses of GenAI, reported by half of the participants. They described using GenAI to learn new programming languages, generate code snippets, and debug or correct errors. In these accounts, GenAI was commonly framed as a helpful and time-saving tool that lowered barriers to coding and debugging, particularly for students with less programming experience. P5 (Medicine, PhD Year 1) described using ChatGPT to learn new programming languages and to make sense of unfamiliar code structures:

> *I have tried to use ChatGPT to help me learn some new programming languages. During this process, I found that ChatGPT may provide some template code for me which, compared with the traditional search agents, provides more efficient clues that help me understand the logic underlying a complex structure of code.*

Despite these efficiency gains, participants reported that AI-generated code could appear syntactically correct but fail to run or produce the intended results.

Domain-specific tasks: Five participants reported using GenAI for domain-specific research tasks requiring specialised expertise, including specialised document and character processing, scientific modelling, and the development of AI-powered software applications. For example, P4 (Dentistry, PhD Year 1) described using AI tools to model protein structures and interactions before conducting wet-lab experiments:

> *I have some AI tools, especially for protein-to-protein communication and interactions. For example, I use AutoDock and I use Alphafold3 to generate the basic structure of a protein such as collagen. For my project, I use AutoDock to simulate how proteins interact with each other.*

Similarly, P11 (Science, PhD Year 3) who mainly focused on bio-images and generative models, described using GenAI to "*help acquire bio-images like microscopy of clean bio image samples, to break the limitation of the hardware*".

Academic writing: GenAI-assisted academic writing was reported by five participants. They described using GenAI to support the drafting of parts of research papers or research proposals, instead of generating complete manuscripts, which they regarded as a form of copying rather than legitimate authorship. P4 (Dentistry, PhD Year 1) articulated this distinction clearly:

> *I think AI can help me to write some parts of articles, but I won't use it to generate my paper from the very start. For example, if AI generates an article and I copy it as my own, I won't do that. I don't think everyone should do this. I think it's just copying. It's not correct.*

P22 (Medicine, PhD Year 1) raised a complementary concern about the loss of human voice in fully AI-generated text:

> *I have a problem if people use AI to write an entire paper. If you read an AI piece, you can tell there's something not quite right. It lacks the human touch, the human interpretation. I like to see a bit of humanity coming through, even in the sciences.*

Editing and proofreading: Editing and proofreading emerged as the most prevalent category of GenAI use, reported by over half of the participants. Participants described relying on GenAI to correct grammar and spelling, refine language and style, and improve readability and coherence of manuscripts, survey questions or other written materials. GenAI was consistently described as a post-writing refinement tool, supporting clarity and presentation without contributing substantively to content development. For non-native English speakers, GenAI was described as particularly valuable for improving linguistic clarity while retaining human control over the final text. P12 (Science, PhD Year 4) reflected on this iterative process:

> *For my thesis writing, I asked AI to modify the text. Since English is not my first language, I find this acceptable. But I still need to modify the AI's version myself. It's an iterative process, often involving several rounds of revision until I'm satisfied.*

Visual content generation and editing: Five participants reported using GenAI to generate or edit visual and multimedia content, including visual design, image and video generation, and image modification. These uses served both creative and communicative functions. For students in design-oriented disciplines, GenAI was integrated into creative workflows to support design exploration and visual production. Other participants reported using GenAI more pragmatically to support academic communication and dissemination, such as preparing visual materials for presentations. P14 (Social Sciences, MPhil Year 1) described this communicative use:

> *I use AI to generate figures or pictures for my slides or posters.*

Administrative and routine tasks: Six participants reported using GenAI to support administrative and routine research-related tasks, including drafting or polishing emails, completing forms, rehearsing interviews or meetings, and formatting citations. These tasks were typically described as low-stakes and repetitive, with GenAI used to reduce administrative burden and free up time for more intellectually demanding research activities. P6 (Business and Economics, PhD Year 3) matched their choice of tool to the perceived demands of the task:

> *If I want to write an email or reply to my professor, I sometimes rely on the university's version of ChatGPT, because it's free and the task doesn't need that much capacity.*

Beyond such drafting tasks, GenAI also supported preparation for high-stakes encounters. P18 (Social Sciences, PhD Year 3) described using AI to rehearse meetings with a busy supervisor:

> *I ask AI to do a role play with me – like, "suppose you're an established professor in my field and I am your research student; what you can do for me?" In real life, my supervisor is super busy, so I usually do a mock interview before I meet my supervisor in person to discuss some research ideas. AI is really efficient for preparing the kind of interview.*

Translation: One participant, P21 (Social Sciences, PhD Year 1), described "*frequently using AI tools to translate Mandarin into English*", highlighting GenAI's role in supporting multilingual research communication.

Conceptualisations of ethical boundaries in GenAI use in research (RQ2)

Our analysis identified five themes that shaped students' conceptualisations of ethically acceptable and unacceptable uses of GenAI in research: human authorship and responsibility, human-centred research, accuracy and reliability of AI output, learning and critical thinking, and institutional rules and disciplinary norms.

Human authorship and intellectual responsibility

This theme captures participants' emphasis on preserving human authorship, intellectual ownership, and responsibility in research activities involving GenAI. Ethical acceptability was primarily framed around whether GenAI use altered the locus of intellectual contribution or decision-making within the research process.

GenAI was widely considered acceptable when used for surface-level or mechanical support, such as language polishing, technical data processing, or routine administrative writing. These uses were seen as enhancing efficiency without altering the substantive intellectual contribution of the researcher.

In contrast, generating research ideas, producing original academic arguments, or even writing the whole paper in place of the researcher were consistently rejected by students, as these practices delegate core scholarly work to AI and thereby violate fundamental norms of authorship and originality. As P21 (Social Sciences, PhD Year 1) noted:

> *As researchers, what we write in our manuscripts is supposed to be original, derived from our research, our data, and our own intellectual effort. For this reason, I do not think AI should be used to produce original content or knowledge.*

Notably, idea development and discussion occupied a conditional space. Participants did not reject all AI involvement in ideation; rather, they accepted brainstorming or exploratory discussion with AI when human researchers retained decision-making authority, critically evaluated AI-generated suggestions, and developed the final ideas through their own scholarly judgement. This distinction reveals an important tension in participants' ethical reasoning. GenAI could be valued as a dialogic aid that stimulates reflection, but it became ethically problematic when it displaced the researcher's own conceptual labour. The boundary between acceptable assistance and unacceptable substitution therefore depended less on the task category itself than on where intellectual agency, ownership, and accountability were located. In this sense, AI-assisted ideation challenged participants to distinguish between being supported in thinking and outsourcing thinking, a distinction that is central to authorship and originality in research.

Human-centred research data

This theme reflects participants' concerns about using GenAI in studies involving human participants, lived experiences, or sensitive qualitative data. Participants expressed strong reservations about uploading human-related or participant data to AI platforms, citing potential risks to privacy, confidentiality, and data security.

Similar strong opposition was expressed regarding the use of GenAI to interpret qualitative, experiential, or human-centred data. These forms of analysis were viewed as requiring human judgement, empathy, and contextual understanding that participants believed AI systems lack. P26 (Arts, PhD Year 2) emphasised that AI cannot meaningfully engage with the lived experiences embedded in such data:

> *When you engage with people whose data you collect and interpret, AI cannot substitute for human experience during the analysis process. AI may help draw general conclusions, but it cannot develop deeper interpretations or relate the data to specific discourses, cultural contexts, or personal and individual experiences.*

Accuracy and reliability of AI output

This theme captures participants' emphasis on the epistemic reliability and quality of AI-generated outputs as a central criterion consideration in their ethical reasoning about whether particular uses of GenAI were acceptable in research. In this study, epistemic reliability refers to the trustworthiness of AI-generated outputs as a basis for scholarly knowledge claims. For participants, this concept distinguished ethical from unethical use by whether such outputs are factually accurate, verifiable against authoritative sources, internally coherent, and aligned with academic quality standards., This issue emerged inductively in participants' discussion of task-specific uses of GenAI, particularly in relation to literature searching where several participants expressed concern about false or fabricated AI-generated references. For example, both P6 (Business and Economics, PhD Year 3) and P28 (Arts, MPhil Year 1) noted that AI systems may produce false or fabricated references, and that relying on such outputs was consistently viewed as unacceptable because it undermined the trustworthiness of academic work and posed risks to research integrity. As a result, participants stressed the importance of human oversight and verification when engaging with AI-generated references or research resources. As P6 (Business and Economics, PhD Year 3) suggested:

> *You have to check out the sources of that information and read through all the papers that they have suggested.*

Extending this reasoning beyond the context of literature review, students evaluated ethical acceptability more broadly based on whether AI outputs met expected academic or professional quality standards, and suggested that ethical reasoning was closely tied to epistemic reliability rather than tool usage per se. As P27 (Arts, PhD Year 1) noted:

> *What really matters is the final output. If the content clearly shows signs of being AI-generated, such as a messy structure, lack of coherence, or obvious inaccuracies, then it should be restricted or prohibited. However, as long as the content is well produced and meets academic or professional standards, I think the use of AI is acceptable, especially given how difficult it has become to distinguish between the two."*

Protecting learning and critical thinking

This theme reflects participants' concern that GenAI use should support, rather than undermine, learning and critical thinking. Their ethical reasoning was closely tied to whether GenAI use, or the deliberate choice not to use it, preserved learners' active cognitive engagement and contributed to meaningful understanding.

Participants generally viewed GenAI as acceptable when it was used to support comprehension, such as explaining concepts or aiding comprehension when reading academic papers. In these tasks, GenAI functioned as a supplementary instructional resource rather than a substitute for independent thinking. However, uses that reduced cognitive effort or engagement with core intellectual tasks were deemed unethical. Importantly, what counted as a "core" intellectual task varied with disciplinary norms,

and the boundary between acceptable and unacceptable GenAI use was drawn accordingly. P22 (Medicine, PhD Year 1) considered the outsourcing of academic writing to AI as a clear case of unacceptable use, on the grounds that producing a written piece of work itself involves substantial cognitive effort and critical thinking:

> *I think it's not acceptable to copy and paste a paper or get it to be paraphrased so it doesn't sound like you're copying the paper and then putting it into your paper. So stuff like that where there is very minimal cognitive use and critical thinking being used to produce a piece of work…the whole point of going to school is…to challenge yourself and learn.*

By contrast, P6 (Business and Economics, PhD Year 3) treated GenAI-assisted coding as acceptable because coding, in their discipline, was a supporting skill rather than a core ability. P6 explained that what mattered was that the researcher "*can run the code themselves and can debug and fix the problems*".

This contrast suggests that participants' ethical reasoning was shaped less by the tool itself than by how a given task was positioned within their disciplinary practice: GenAI assistance was more acceptable when applied to supporting or procedural tasks, but more problematic when it displaced activities understood as central to scholarly formation or intellectual contribution.

These concerns were especially salient in assessment contexts, including confirmation of candidature and thesis submission, where assessment is intended to directly evaluate students' understanding or research capabilities. The use of GenAI in such contexts was viewed as potentially compromising assessment validity by preventing accurate evaluation of learning outcomes. However, participants considered GenAI use acceptable in assessment settings where learning outcomes could still be reliably demonstrated and verified. To address these challenges, participants suggested shifting from written assessments to oral or face-to-face formats, which were perceived as less susceptible to AI-mediated assistance. As P12 (Science, PhD Year 4) explained:

> *My suggestion is to change the format of assessment. For example, written examinations or homework could be replaced with oral examinations. In face-to-face interactions, it is essentially impossible to rely on AI, and it becomes much easier to judge whether students truly understand the concepts and possess the necessary ability to solve problems.*

Institutional rules and disciplinary norms

This theme captures students' reliance on institutional policies and disciplinary expectations as reference points for judging the ethical acceptability of GenAI use in research.

One participant indicated that GenAI use was considered ethically acceptable when it was explicitly permitted by university or departmental policies, or when it aligned with established guidelines. At the same time, students recognised that disciplinary norms surrounding GenAI use are evolving. For example, P7 (Business and Economics, PhD Year 2) described a shift in attitudes towards AI-assisted idea generation within their discipline:

A paper published by the best journal of our area is discussing about how to train the AI to generate good idea. I was very surprised and are actually trying to use AI to generate ideas.

Ethical concerns regarding GenAI use in research (RQ3)

Analysis of students' responses revealed a multifaceted set of ethical concerns regarding the use of GenAI in research.

Ethical concerns about research data and ideas

Participants raised concerns about data integrity, data privacy, data security and research data leakage associated with the use of GenAI in research. Research data often contain sensitive or personal information, particularly in studies involving human participants. When data are uploaded to GenAI systems, there is a risk that the original data may be altered or contaminated, that sensitive or personal data may be stored or analysed by the system, or that locally stored data may be damaged. As P28 (Arts, MPhil Year 1) explained:

> *Many historical documents, such as records of enterprise development, family correspondence, or personal letters, contain detailed personal and familial information. If such documents are captured (e.g. via screenshots), digitised, or converted into PDFs using AI tools, there is a risk that the information may be processed, stored, or incorporated into AI systems beyond the researcher's control. This raises serious ethical concerns regarding data privacy, consent, and information leakage in the use of AI for historical research.*

In addition to data-related risks, participants also voiced apprehension about research idea leakage, particularly when unpublished ideas, plans, or drafts are shared with GenAI systems. They suggested avoiding the upload of critical data or novel ideas to GenAI tools and instead storing such materials securely and locally to safeguard the originality and integrity of ongoing research.

Ethical concerns about AI output quality

Participants' ethical concerns were strongly shaped by perceptions of the quality and reliability of AI-generated outputs. The most frequently cited concerns are hallucinated facts, fabricated references, or misleading content. As P28 (Arts, MPhil Year 1) described:

> *GenAI systems sometimes struggle to accurately identify the correct journal articles, book chapters, or publishers. When repeatedly prompted for precise bibliographic details, they may instead generate fabricated or inaccurate references.*

Beyond factual inaccuracies, participants raised concerns about epistemic bias, noting that AI systems may privilege dominant or mainstream perspectives while marginalising less visible scholars or alternative viewpoints when doing literature reviews. Others pointed to an erosion of scholarly quality, describing AI-generated text as repetitive, superficial, or lacking conceptual depth. Concerns about AI reliability further reinforced scepticism about using GenAI for research-critical tasks, with some students explicitly questioning whether current AI tools are sufficiently stable or dependable for rigorous academic work.

Ethical impacts on research quality and integrity

Another major theme concerned the perceived impact of GenAI use on research integrity and scholarly norms. Students identified GenAI as posing a threat to research replicability, citing the non-deterministic and opaque nature of GenAI systems as making research processes and outcomes difficult to reproduce. P6 (Business and Economics, PhD Year 3) elaborated on this concern in the context of empirical work that uses GenAI to generate measures or variables:

> *One problem I face when reading some AI related papers in my own discipline is that people use AI to generate some measures, and when they use different models from different providers, the results are quite different. The replication problem is a very important issue in research. We have to ensure that the results obtained using one model can be replicated by the other researchers. But if we use AI, it's often difficult to replicate others' work, compared with using Python code.*

Concerns about research originality were also evident. Participants questioned whether GenAI can produce genuinely original ideas, given that AI systems primarily reproduce or recombine content from their training data. P5 (Medicine, PhD Year 1) raised this question directly:

> *I would like to know whether AI can truly generate original or novel ideas. I think the most important things in a PhD thesis are originality and novelty. If AI cannot achieve that, then AI can only serve as a tool that smooths or supports our daily work. So, using the AI or not may not affect the originality and novelty of the thesis. If the originality can only come from humans, I'm not sure whether using AI affects the core values of postgraduate studies and PhD thesis.*

Building on this concern, participants noted that the line between AI-generated and human-generated ideas is difficult to draw, complicating judgements about plagiarism. P28 (Arts, MPhil Year 1) articulated this in the context of historical research, where "*the way you understand, or the way you organise your sources into a coherent narrative or coherent idea*" is itself a form of original contribution:

> *If you simply transplant the ideas from the generative AI as your own idea, that can be a form of plagiarism. But it's very difficult to identify the boundary between plagiarism and whether your idea is from generative AI or elsewhere.*

These concerns about originality extended into questions of authorship and accountability, particularly when AI-generated content is incorporated into formal scholarly outputs. P13 (Social Sciences, PhD Year 2) raised this in relation to journal submissions:

> *In teaching, I've seen people just blindly copy and paste what AI has given them and submit that, which from a research ethics standpoint, I think raises a lot of issues, especially with journal submissions and norms around authorship, given that AI can't be listed as an author.*

P20 (Education, PhD Year 3) offered a more relational view of authorship, drawing analogies between AI-assisted research and the everyday intellectual exchanges through which scholarly ideas develop:

> *Many of us researchers engage with our counterparts, we discuss ideas, and many times during those conversations we come up with an idea – a stroke of genius or epiphany. Nothing as researchers is 100% original – knowledge is constructed based on previous knowledge, and through our interactions we develop deeper insight.*

P20 argued that legitimate AI use, like legitimate intellectual conversation, depends on the researcher's capacity to articulate their own ideas, take ownership of them, and process AI-derived input critically rather than accept it uncritically.

Ethical impacts on individuals and institutions

Beyond research outputs, participants articulated concerns about the broader ethical consequences of GenAI use for individual researchers and institutions. A recurrent concern involved over-reliance on AI, which students warned could diminish core scholarly capacities such as critical thinking and creativity.

P24 (Business and Economics, PhD Year 2) argued that AI cannot replicate the disciplinary "taste" that drives original research:

> *Researchers have field expertise and their own intellectual taste. You don't know everything, but you have your field knowledge and a specific taste in that area, which lets you test what may seem like an unrealistic hypothesis. AI can give you the whole picture, but it cannot give you anything that seems crazy or unrealistic. I think that is the core part of scientific research – proposing something that doesn't yet seem realistic. So if we rely too much on AI, we will reduce our creativity.*

P23 (Engineering, PhD Year 4) described peers accepting AI outputs without scrutiny:

> *Many of my friends always use AI tools such as ChatGPT or DeepSeek. If I ask them something, they will open DeepSeek and type in my question. But many times the response is not true, and there is unrealistic and unreliable information. Most of my friends believe it and think it's true. I think it's a very big problem.*

P13 (Social Sciences, PhD Year 2) referenced a recent news incident to illustrate how AI misuse by individuals can damage the reputation of their institutions:

> *I saw a news article about someone at another university who graduated and then bragged that they had used AI to write their whole degree or thesis. People were saying that person should have their degree revoked. There needs to be a balance between encouraging students to access these resources to facilitate academic progress, without also damaging the reputation of the school — like what happened in that news story.*

Institutional support for responsible GenAI use in research (RQ4)

Students identified multiple forms of institutional support as necessary for enabling the responsible GenAI use in research. Over one-third of participants emphasised financial support, particularly access to paid AI tools or coverage of subscription fees. Given the difficulties of affording subscription costs, students often defaulted to free AI tools with limited capabilities. If such costs continue to be borne individually, access to higher-quality tools risks becoming uneven across the student body, potentially affecting research opportunities. Students also called for operational guidance to support effective

use of GenAI. P28 (Arts, MPhil Year 1) requested for curated lists of credible AI tools, and P25 (Architecture, PhD Year 2) requested concrete use cases and prompt templates tailored to specific academic disciplines.

Participants also identified institutional policies and guidelines as a critical form of support. While many supported clearly articulated policy content that explicitly defines acceptable and unacceptable AI use, they emphasised that such policies should encourage responsible use rather than impose blanket restrictions. P19 (Social Sciences, PhD Year 2) referred to their personal experience as students in courses with restrictive AI policies such as requirements to report all AI use:

> *I personally embrace AI rather than taking a very tight hand in handling AI policy, because previously I experienced courses that had a very strong hand when it comes to AI usage. They ask you to report everything. I think these kind of ways of handling it would not work because there are always ways that people can get around them. It would be better to encourage people to use it responsibly rather than putting a lot of restrictions and guidelines that force everyone to follow.*

Beyond infrastructure and policy, students strongly emphasised the need for capacity-building initiatives to develop researchers' AI literacy across its knowledge, practical, evaluative, and ethical dimensions. P20 (Education, PhD Year 3) argued that responsible use depends on first identifying AI's specific risks, drawing an analogy with everyday tools:

> *Part of the learning is to know what AI can and cannot do before we use something responsibly. We need to understand what the dangers are. When we go buy a chair, for instance, nobody asks us to sign a disclosure to use the chair safely, because we know there is no inherent danger. When we buy a gun, you need to start asking questions – there is an inherent danger, and you know what the danger is. So what are the dangers of using AI? Every time we ask this question, people don't have answers, and since they don't have answers, they're puzzled when we come to the policy.*

P20's argument extends beyond individual capacity-building: the absence of a clear understanding of AI's risks impedes the development of meaningful policy, positioning capacity-building and policy as interdependent rather than separable forms of institutional support.

Participants also commented on the training currently available. P19 (Social Sciences, PhD Year 2) noted that the host university's graduate school had provided some training sessions on AI for postgraduate students. However, more participants felt that the existing provision was insufficient for research-specific needs. P13 (Social Sciences, PhD Year 2) observed that "*there isn't a lot of guidance relevant to research and writing manuscripts*" and called for "*more training on how we can use AI effectively or use it in a more responsible manner than what we've currently got*". P16 (Education, PhD Year 3) called for more structured provision, suggesting that "the graduate school should have a compulsory course on AI, just like thesis writing, teaching students how to use AI, for example, what kind of inputs can be inserted, what questions you should ask, how to ask the WH-question, how to provide context, and what kind of use is encouraged by the graduate school".

Beyond institutional training, some participants reported receiving informal guidance from their supervisors, which shaped how they engaged with GenAI in their research. Supervisor messages were not uniform. P4 (Dentistry, PhD Year 1) was actively encouraged to use GenAI for specific purposes: "Actually, my supervisor encouraged me to use AI for grammar check before I submit my final paper to him". In contrast, P7 (Business and Economics, PhD Year 2) was warned against using GenAI in ways that risked exposing original research data: "My supervisor forbids me from uploading the texts I have written to GPT. She is afraid of data privacy and data leaking. The data here is about our original ideas". These contrasting accounts suggest that supervisors are already informal but variable mediators of responsible GenAI use, with the messages students receive depending on individual supervisor views and disciplinary contexts.

To develop practical and implementable AI policies, and to support their consistent adoption, participants stressed the need for transparent communication among faculty members, supervisors, and students, as well as regular review and updating of policies to reflect evolving research practices.

Discussion

Student researchers' enactment of AI literacy in research

This study adopted Ng et al.'s (2021) four-dimensional AI literacy framework to examine how PGR students enact AI literacy in research practice. Findings across the four research questions can be read as situated expressions of each dimension, indicating both the capacities students currently bring to GenAI use and the dimensions that may require further development.

"Know & understand AI" was reflected in students' awareness of GenAI's capabilities, limitations, and risks. Across RQ1, participants demonstrated working knowledge of what specific tools could do and where their boundaries lay – for instance, recognising that AI-generated code may appear syntactically correct but fail to run, that AI-generated ideas could be inaccurate or misleading, and that AI explanations supported introductory but not in-depth understanding. The concerns surfaced in RQ3, particularly around AI's non-deterministic behaviour and limited capacity for genuine originality, further illustrate this dimension.

"Use & apply AI" was reflected in the diverse, discipline-specific applications participants reported across the research workflow. Findings under RQ1 demonstrated that students used GenAI for ideation, literature review, explanation, data processing, programming, domain-specific tasks, academic writing, editing, visual content generation, administration, and translation. Crucially, this dimension was enacted not as uniform reliance on GenAI but as calibrated, task-sensitive use – students matched tools to tasks based on perceived stakes, intellectual demands, and disciplinary norms, with heavier use in procedural or low-stakes work and more cautious engagement in tasks central to scholarly contribution.

"Evaluate and create AI" was reflected in students' nuanced, self-defined boundaries between acceptable and unacceptable use, surfaced through RQ2. Participants did not treat GenAI's outputs as authoritative; they evaluated them against their own judgement, domain knowledge, and emerging ethical principles. This evaluative work was particularly visible in their distinctions between drafting and full generation in academic writing, between AI-assisted ideation and AI-derived ideas in research design, and

between linguistic refinement and substantive intellectual contribution in editing. The boundary judgements participants articulated were not derived from external rules but constructed through ongoing reflection on their own research practice.

"AI Ethics" was reflected in the substantive concerns participants held about responsible GenAI use, surfaced primarily through RQ3. These spanned methodological concerns (replicability), conceptual concerns (originality and plagiarism), normative concerns (authorship and accountability), capacity concerns (over-reliance and creativity loss), and institutional concerns (reputational risks). The breadth and specificity of these concerns indicates that participants were already engaged in active ethical reasoning about AI in research, rather than treating it as a value-neutral technology.

Together, these four dimensions reveal that PGR students are not passive recipients of GenAI tools but active interpreters and negotiators of AI in research. They exercised knowledge, application, evaluation, and ethical reasoning in ways shaped by their disciplines, research stages, and personal judgement. At the same time, RQ4 findings indicate that students have developed this AI literacy largely on their own, through their own judgement and informal learning from peers, with little structured guidance or training from their institutions. This combination, i.e., students who are capable but lack institutional support, gives us the data-based justification for proposing the task-sensitive, researcher-oriented guidelines presented in the next section.

Contextual influences on students' practices and ethical reasoning

The findings suggest that participants' practices and views regarding GenAI use in research were shaped by several immediate contextual influences. Institutional policies provided reference points for judging whether particular uses of GenAI were acceptable, although participants often perceived existing guidance as either insufficiently research-specific or overly restrictive. Disciplinary norms also shaped participants' reasoning by influencing what counted as core intellectual work, acceptable evidence, appropriate methods, and legitimate authorship within a field. At the same time, participants' own AI literacy affected how they interpreted GenAI's capabilities, limitations, and risks, including whether they could evaluate AI-generated outputs critically and decide when human oversight was required. Supervisor messages further shaped students' practices, but these messages were uneven: some supervisors encouraged limited uses such as grammar checking, while others warned students against uploading original data or ideas because of privacy and confidentiality concerns.

Participants' practices and views regarding GenAI use in research may also have been influenced by broader contextual factors, including public narratives about AI misuse and academic pressures to produce high-quality, original, and publishable research. For example, public discussions of students or researchers using AI to produce whole assignments, theses, or publications may have heightened participants' concerns about academic misconduct, reputational risk, and the need to preserve human authorship. Academic misconduct, including fabrication, falsification, plagiarism, or misleading reporting, is often met with strong institutional and public criticism. Researchers are expected to take responsibility for scholarly work published under their names, regardless of whether problematic content originates from themselves, collaborators, or AI systems.

The potential consequences of misconduct, including reputational damage, institutional investigation, publication retraction, loss of trust from journals or conferences, and harm to students, collaborators, or affiliated institutions, may encourage researchers to approach GenAI use cautiously. This wider academic and reputational environment helps explain why participants were particularly concerned about using GenAI for tasks closely tied to authorship, originality, evidence, and research claims, such as idea generation, data interpretation, and manuscript writing. Their ethical reasoning should therefore be understood not only as individual reflection on right or wrong use, but also as situated within broader institutional, disciplinary, and public expectations about responsible scholarship.

Implications for doctoral training and supervision

The findings suggest that responsible GenAI use in research requires not only policies, but also more structured doctoral training that develops AI literacy as a research competence. Rather than remaining at the level of abstract principles, such training could use concrete, discipline-sensitive examples or case studies to facilitate discussion and help PGR students understand GenAI's capabilities and limitations, apply it appropriately across research tasks, evaluate AI-generated outputs, and reason through ethical concerns such as originality, privacy, reliability, authorship, and over-reliance.

The findings also indicate that supervisors already shape students' GenAI practices, but often in uneven and informal ways. Supervisor-student dialogue should therefore become a more explicit part of responsible GenAI support, particularly when students are making decisions about core intellectual work such as research design, data analysis and academic writing. This is especially important because acceptable GenAI use often depends on disciplinary norms and the role of a task within a specific research project. Supervisors are usually more accessible to students than institutional policy actors and are better positioned to interpret contextual expectations. They can help students distinguish between appropriate assistance, such as language refinement and coding support, and inappropriate delegation of scholarly work, such as outsourcing ideation or data interpretation to GenAI.

Task-sensitive guidelines for responsible GenAI use and AI literacy development in research

Building on the findings, this section presents a set of guidelines designed to support PGR students in using GenAI responsibly across the diverse tasks of research practice while simultaneously developing their AI literacy over time. The guidelines presented in Table 4 are organised as a two-layered framework: the first layer corresponds to common research tasks identified in this study, and the second layer engages each of the four AI literacy dimensions – "Know & Understand AI", "Use & Apply AI", "Evaluate and Create AI", and "AI Ethics" – within each task. Rather than prescribing fixed rules, the guidelines articulate the knowledge, considerations, and practical advice that responsible GenAI use requires in each task, with the aim of cultivating the situated judgement researchers need to navigate complex and evolving research situations. In doing so, the guidelines are intended to function not only as an immediate practical resource but also as a developmental scaffold for the sustained growth of researchers' AI literacy.

Table 4 Task-sensitive guidelines for responsible GenAI use and AI literacy development in research, informed by participants' practices and Ng et al.'s (2021) AI literacy framework

Research task	Know & understand AI	Use & apply AI	Evaluate & create AI	AI ethics
Idea generation and brainstorming	(i) GenAI-generated ideas may be inaccurate, generic, misleading, or derivative; (ii) GenAI outputs may recombine existing content rather than produce genuinely original insights.	Generate, discuss, refine, or polish emerging research ideas, especially at the early exploratory stage of research design.	(i) Human researchers should critically evaluate whether AI-generated ideas are conceptually meaningful, relevant to the research problem, and sufficiently developed through their own scholarly judgment; (ii) Human researchers should retain decision-making authority and intellectual ownership over the final research ideas;	Ethical considerations include leakage of unpublished ideas, plans, or drafts; uncritical appropriation of AI-generated ideas; plagiarism; blurred authorship; and presenting AI-dominated work as human-authored, thereby obscuring accountability to supervisors, reviewers, editors, or examiners.
Literature review	(i) GenAI may produce false, incomplete, outdated, or fabricated references; (ii) GenAI may fail to identify correct journal articles, book chapters, publishers, or bibliographic details.	(i) Search for relevant literature, gain an overview of a topic; (ii) Summarise existing studies, and support initial understanding of prior research.	(i) Researchers should verify all AI-generated references and summaries against authoritative databases and original sources; (ii) They should also assess whether AI-generated literature reviews are sufficiently comprehensive, balanced, and aligned with the research question.	Ethical considerations include fabricated citations, misrepresentation of prior work, over-reliance on AI-generated summaries, and bias toward dominant or mainstream perspectives while marginalising less visible scholars, or alternative viewpoints.
Explanation	GenAI explanations may simplify, distort, or misrepresent complex theories, mathematical derivations, methods, or disciplinary concepts.	(i) Explain unfamiliar concepts; (ii) Clarify mathematical derivations; (iii) Support understanding of research methods.	(i) Researchers should compare AI-generated explanations with textbooks, methodological sources, supervisor feedback, or peer-reviewed literature; (ii) AI support should enhance understanding rather than replace researchers' own conceptual learning.	Over-reliance may lead to superficial understanding, hidden knowledge gaps, or skill degradation, especially if students accept explanations without verification.
Data processing and preparation	GenAI may misread, alter, contaminate, or misclassify data, especially when handling complex, sensitive, or context-dependent datasets.	(i) Support preprocessing or post-processing of raw data, data cleaning, formatting, categorisation; (ii) Support data labelling, such as labelling video or textual data collected for research.	(ii) Researchers should evaluate the accuracy, consistency, and traceability of AI-assisted data processing; (ii) They should maintain human oversight, document AI involvement, and check whether AI outputs preserve the integrity of the original data.	Ethical considerations include unauthorised sharing of sensitive or personal information, contamination of original data, storage or processing of confidential data by third-party systems, and potential damage to locally stored data.

Table 4 (continued)

Research task	Know & understand AI	Use & apply AI	Evaluate & create AI	AI ethics
Programming and coding	AI-generated code may appear syntactically correct but fail to run, produce incorrect results, introduce hidden errors, or be unsuitable for the intended analytical task.	(i) Learn programming languages; (ii) Generate code snippets, debug errors; (iii) Support coding for data analysis.	(i) Researchers should test, debug, and understand AI-generated code before use; (ii) Responsible use requires active engagement, reflection, and verification rather than passive adoption of AI-produced solutions.	Ethical considerations include using code that produces invalid analyses, failing to understand or explain one's own computational procedures, reproducing biased or insecure code, and weakening researchers' programming competence through over-reliance.
Academic writing	GenAI-generated academic text may be repetitive, superficial, inaccurate, poorly grounded, or lacking conceptual depth and disciplinary nuance.	Support the drafting of selected parts of research papers, proposals, abstracts, summaries, or other academic texts.	(i) Researchers should evaluate whether AI-generated writing accurately represents their argument, evidence, methods, and contribution; (ii) AI should support expression, organisation, or drafting, but should not replace the researcher's substantive intellectual contribution.	Ethical considerations include blurred authorship, loss of originality, misrepresentation of contribution, plagiarism, excessive dependence on AI-generated text, and delegation of core scholarly work to AI in ways that violate academic integrity.
Editing and proofreading	(i) Language improvement does not necessarily improve the accuracy, originality, or scholarly quality of the underlying argument; (ii) GenAI may also change meaning while editing.	(i) Correct grammar and spelling; (ii) Refine language and style; (iii) Improve readability and coherence; (iv) Polish survey questions, or other written materials.	(i) Researchers should check whether AI-edited text preserves the intended meaning, disciplinary terminology, methodological precision, and authorial voice; (ii) Editing should enhance communication without altering the substantive intellectual contribution.	Ethical considerations include undisclosed substantial rewriting, distortion of meaning, excessive dependence on AI for scholarly expression, and inequities around language support and academic authorship norms.
Visual content generation and editing	AI-generated or AI-edited visuals may be inaccurate, misleading, fabricated, biased, or inappropriate for representing research evidence.	Generate or edit visual and multimedia content, including images, diagrams, videos, presentation materials, posters, and other forms of academic communication and dissemination.	(i) Researchers should evaluate whether AI-generated visuals accurately represent the underlying data, concepts, or research message; (ii) Visual outputs should support communication without fabricating evidence or exaggerating claims.	Ethical considerations include misleading visual representation, fabrication or manipulation of research evidence, copyright and attribution issues, inappropriate use of AI-generated visual content.

The guidelines presented in Table 4 were derived from a structured synthesis of the study's empirical findings and Ng et al.'s (2021) four-dimensional AI literacy framework. The derivation process drew on each of the four research questions in distinct ways. Findings from RQ1 identified the diverse research tasks across which PGR students reported using GenAI; these tasks formed the first organisational layer of the guidelines, and informed the "Use & Apply AI" dimension. Findings from both RQ1 and RQ2 informed the "Evaluate & Create AI" dimension, drawing on participants' descriptions of how they used GenAI within each task and on the boundary judgements through which

they distinguished acceptable from unacceptable use. Findings from RQ1, RQ2, and RQ3 informed the "Know & Understand AI" and "AI Ethics" dimensions, capturing participants' awareness of GenAI's task-specific limitations and the ethical risks they identified across replicability, originality, plagiarism, authorship, and over-reliance. Findings from RQ4 shaped the guidelines' overall orientation: rather than imposing prohibitions, the guidelines articulate the knowledge, considerations, and practical recommendations researchers need to exercise responsible judgement themselves.

The synthesis was guided by two principles. First, the guidelines were grounded in responsible practices participants themselves articulated, rather than in external prescription, recognising that PGR students are already enacting AI literacy in their research. Second, the guidelines were extended where empirical practices were uneven or incomplete, drawing on the framework to identify dimensions of AI literacy that participants engaged with implicitly but did not always articulate explicitly.

Table 4 is intended as an illustrative scaffold rather than an exhaustive or definitive set of rules. The research tasks listed correspond to those reported by participants in this study and may not capture every task or sub-task encountered across different disciplines, methodologies, or research stages. Likewise, the considerations articulated under each AI literacy dimension reflect current GenAI capabilities and the ethical concerns surfaced in our data, both of which will continue to evolve. Disciplines, supervisors, and institutions are expected to adapt, extend, and revise the scaffold to reflect their own norms, regulatory contexts, and the evolving GenAI landscape, ideally through iterative dialogue with the researchers it is intended to support.

Limitations and future directions

Several limitations of this study should be acknowledged. First, participants were not purposively sampled to ensure representation across faculties, resulting in an uneven disciplinary distribution. For example, the sample included a higher proportion of students from the social sciences and limited perspectives from disciplines such as Law. As disciplinary norms, research practices, and ethical expectations vary substantially, future studies should involve participants from a broader and more balanced range of disciplines to examine disciplinary differences in GenAI use and ethical reasoning more systematically.

Second, this study did not systematically examine departmental-level GenAI guidelines, nor did it directly investigate supervisor perspectives or supervisor-student dialogue around GenAI. While several participants referred in passing to informal guidance received from their supervisors, the study's design centred on PGR students' own practices and reasoning rather than on the institutional and relational structures around them. Future research should examine departmental variation in GenAI guidance, incorporate supervisor perspectives, and investigate how supervisor-student conversations shape the development of AI literacy in research.

Third, this study did not systematically examine variation by discipline, stage of study, familiarity with GenAI, or type of research task. Although participants were encouraged to draw on their own research experiences and disciplinary contexts, and although some task-specific and discipline-related insights emerged inductively, the uneven distribution of participants across disciplines and study stages limits the basis for systematic comparative claims. Future research should examine such differences more systematically,

for example through purposive sampling across disciplines, larger-scale surveys, or comparative qualitative designs.

Fourth, the reliance on self-reported interview data may introduce social desirability or response bias. Participants may have underreported certain uses of GenAI due to concerns about academic integrity, stigma associated with "cheating", or fear of potential consequences. Future studies could complement self-reported accounts with observational data, system logs, or anonymous surveys to triangulate findings and reduce such biases.

Fifth, two focus groups were documented through written notes rather than audio recordings at participants' request. The notes were managed by two researchers and consolidated after each session to improve accuracy and retain participants' key points, examples, and main discussion threads. Nevertheless, written notes provide less detail than verbatim transcripts and may not fully capture the nuance of participants' wording or group interaction. Future studies could seek participants' consent for audio recording wherever possible. When audio recording is not permitted, researchers could consider using alternatives, such as automatic note-taking tools or real-time transcription tools, subject to participants' consent and relevant ethical requirements.

Finally, the rapid pace of development in generative AI technologies means that specific tools, capabilities, and usage practices identified in this study may evolve quickly. While the core ethical concerns and reasoning patterns identified here are likely to remain relevant, future research should continue to revisit and update empirical understandings of GenAI use to keep pace with technological change.

Conclusion

The rapid integration of GenAI into research has outpaced the development of guidance that is specific, practical, and grounded in researchers' lived experience, leaving postgraduate researchers, among the most active adopters of GenAI tools, to navigate complex ethical decisions without adequate institutional support. To address this gap, this study examined how PGR students enact AI literacy in research through seven focus groups with 28 participants, drawing on Ng et al.'s (2021) four-dimensional framework to analyse their practices, ethical reasoning, concerns, and institutional support needs. The findings revealed that PGR students are not passive adopters of GenAI but active users who enact AI literacy across all four of its dimensions in their research practice. They demonstrated "Know & Understand AI" through awareness of GenAI's task-specific capabilities and limitations, "Use & Apply AI" through diverse, discipline-sensitive applications across the research workflow, "Evaluate & Create AI" through nuanced boundary judgements between assistance and substitution, and "AI Ethics" through reasoned concern about accuracy, authorship, originality, data integrity, and the development of scholarly competence. Building on these insights, the study proposed task-sensitive guidelines that engage all four AI literacy dimensions across diverse research tasks, designed to function not only as a practical resource for responsible GenAI use but also as a developmental scaffold for the sustained growth of researchers' AI literacy.

The contribution of this work is therefore both empirical and theoretical. Empirically, it provided a detailed account of how PGR students enact AI literacy in research. Theoretically, it positioned AI literacy as a situated capacity that researchers develop through ongoing engagement with GenAI's evolving capabilities and limitations, rather than a

static set of competencies to be acquired. As GenAI continues to evolve and embed itself in scholarly work, guidance that moves beyond binary rules toward nuanced, practice-oriented support, anchored in researchers' own reasoning and developmental needs, is essential for enabling ethical and sustainable research practices.

Appendix A: Interview questions

Research questions	Main questions	Follow-up questions
RQ1: How do PGR students use GenAI in research?	Do you use GenAI tools? What purposes do you use GenAI tools for? What GenAI tools do you use for research? What research areas or purposes do you use GenAI for?	What's the reason why you don't use GenAI? If you intend to use in the future, what do you think you might use it for?
RQ2: How do PGR students conceptualise ethically acceptable and unacceptable uses of GenAI in research?	In your opinion, what situations do you think it is acceptable for research postgraduate students to use GenAI? In your opinion, what situations do you think it is not acceptable for research postgraduate students to use GenAI?	
RQ3: What ethical concerns do PGR students hold regarding the use of GenAI in research?	What are your biggest concerns about using GenAI in research in your discipline?	Are there any unique risks in your discipline? Have you personally encountered issues with GenAI in your research?
RQ4: What forms of institutional support do PGR students believe are necessary to enable responsible use of GenAI in research?	Regarding responsible GenAI use for all PGR students, what sort of guidance do you want to see/have?	In your faculty/department, is there any support for you with using GenAI? What do you think is necessary to include in a responsible GenAI use policy for PGR students and research?

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W.D. contributed to conceptualisation, methodology, data analysis, and writing—original draft. C.C. contributed to conceptualiaation, methodology, supervision, and writing—review & editing. All authors read and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests
The authors have no relevant financial or non-financial interests to disclose. The authors declare no competing interests.

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